

DEVELOPEMENT OF A WEARABLE SENSOR SYSTEM FOR EARLY DETECTION OF MUSCLE FATIGUE IN ATHLETES

A PROJECT REPORT

Submitted by

ABDUL MAHATHEER M 8208E22BML301

MOHAMED ARSADH A 8208E22BML303

*in partial fulfillment for the award of the
degree of*

**BACHELOR OF ENGINEERING
IN
BIOMEDICAL ENGINEERING**



**E.G.S. PILLAY ENGINEERING COLLEGE, NAGAPATTINAM
ANNA UNIVERSITY: CHENNAI 600 025**

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NAGAPATTINAM

BONAFIDE CERTIFICATE

Certified that this project report **“DEVELOPEMENT OF A WEARABLE SENSOR SYSTEM FOR EARLY DETECTION OF MUSCLE FATIGUE IN ATHLETES”** is the Bonafide work of **ABDUL MAHATHEER M (8208E22BML301), MOHAMED ARSADH A (8208E22BML303)** who carried out the project work under my supervision.

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ACKNOWLEDGEMENT

It is indeed my responsibility to express my heartfelt gratitude to our founder Chairman Late **Chevalier Dr. G.S. PILLAY**. It is my pleasant duty to express gratitude to our chairman **Mrs. JOTHIMANI G. S. PILLAY**, the paramount of our prestigious institution and academic advisor **Dr. S. PARAMESVARAN, M.Com., FCCA (LONDON)** for his patronage.

I wish to extend our sincere thanks to our beloved secretary **Shri. S. SENTHIL KUMAR**, and to our beloved joint secretary **Shri. S. SANKAR GANESH** and our respected Principal **Dr. S. CHINNADURAI, M.E., Ph.D.**, for having provided me with the necessary facilities to carry out this project work.

I am very much honored and would like to profoundly express deep gratitude to our Head of the Department **Dr. S. CHITRA, M.E., Ph.D.**, for his expert, sincere, and valuable guidance. I also would like to express my sincere thanks and gratitude to our project coordinator **Dr. R. VENKATESAN, M.Tech., Ph.D.**, for his full support and motivation.

I express thanks to my project guide **MRS. S. SUGANYA, M.E.**, our class in charge **Dr. C. USHADEVI, M.E., Ph.D.**, for their wide knowledge, serious research attitude, and enthusiasm for work that made it possible to complete this project at most satisfaction of our superiors in a stipulated period.

I express hearty render and thanks to all the staff members and technical assistants of the Department of **BIOMEDICAL ENGINEERING** for their timely assistance. Our humble gratitude and thanks to our parents and friends who have supported and helped me to complete the project work.

ABSTRACT

Muscle fatigue is a critical factor that affects athletic performance, physical endurance, and the risk of injury during prolonged or repetitive activities. Early detection of fatigue is essential to optimize performance and prevent musculoskeletal damage. This project presents the design and implementation of a wearable muscle fatigue detection system based on surface Electromyography (sEMG). The system captures bioelectrical signals generated by muscle activity using surface electrodes and processes them in real time to analyze fatigue conditions.

The acquired EMG signals are first conditioned using an analog front-end to remove noise and improve signal quality. The conditioned signals are then digitized using a microcontroller and processed using digital signal processing techniques. Key features such as Root Mean Square (RMS) and Median Frequency (MDF) are extracted from the signal to evaluate muscle activity and fatigue levels. While RMS reflects the intensity of muscle activation, MDF provides a reliable indication of fatigue by analyzing changes in the frequency spectrum of the signal. A fatigue detection algorithm is implemented to monitor the decreasing trend of MDF over time and generate alerts when fatigue is detected.

The system utilizes a microcontroller with built-in wireless communication capabilities to enable real-time data transmission to external devices for monitoring and analysis. The experimental results demonstrate that RMS values increase with muscle effort, whereas MDF values decrease as fatigue develops, confirming the effectiveness of the proposed method. The system provides a compact, cost-effective, and user-friendly solution for continuous fatigue monitoring.

This project has significant applications in sports training, rehabilitation, and healthcare, where real-time monitoring of muscle condition can improve performance, enhance recovery, and reduce the risk of injuries. Future enhancements may include the integration of artificial intelligence and multi-sensor systems to further improve accuracy and functionality.

Experimental observations demonstrate a clear increase in RMS values and a corresponding decrease in MDF values during fatigue conditions, validating the effectiveness of the approach. The system's low power consumption and wireless communication capabilities make it highly practical for wearable applications. Furthermore, the system has potential applications in sports, rehabilitation, and healthcare, with future enhancements aimed at incorporating artificial intelligence and multi-sensor integration for improved performance.

Keywords: Surface Electromyography (sEMG), Muscle Fatigue Detection, Wearable Technology, Digital Signal Processing (DSP), Root Mean Square (RMS), Median Frequency (MDF), Fast Fourier Transform (FFT), Real-Time Monitoring

TABLE OF CONTENTS

CHAPTER

TITLE

ACKNOWLEDGEMENT

ABSTRACT

LIST OF FIGURES

LIST OF ABBREVIATIONS

1

INTRODUCTION

1.1 OVERVIEW OF SPORTS BIOMECHANICS

1.2 PROBLEM STATEMENT:

1.3 OBJECTIVES OF THE PROPOSED SYSTEM

1.4 SIGNIFICANCE OF WEARABLE TECHNOLOGY IN ATHLETICS

1.5 THESIS/REPORT ORGANIZATION

2

LITERATURE REVIEW

2.1 PHYSIOLOGY OF MUSCLE CONTRACTION AND FATIGUE

2.2 REVIEW OF SURFACE ELECTROMYOGRAPHY (SEMG)

2.3 ANALYSIS OF CURRENT WEARABLE FATIGUE MONITORS

2.4 COMPARATIVE STUDY OF SIGNAL PROCESSING TECHNIQUES

2.5 SUMMARY OF GAPS IN EXISTING RESEARCH

3

SYSTEM ARCHITECTURE & DESIGN

3.1 HIGH-LEVEL SYSTEM BLOCK DIAGRAM

3.2 HARDWARE COMPONENTS

	3.2.1 SEMG ELECTRODES: WET VS. DRY SELECTION
	3.2.2 ANALOG FRONT-END (AFE) AND SIGNAL CONDITIONING
	3.2.3 MICROCONTROLLER SELECTION (ESP32/ARM)
	3.3 WIRELESS COMMUNICATION PROTOCOL (BLE/WI-FI)
	3.4 POWER MANAGEMENT FOR WEARABLE DEVICES
4	DIGITAL SIGNAL PROCESSING FRAMEWORK
	4.1 SIGNAL ACQUISITION AND SAMPLING RATES
	4.2 PRE-PROCESSING AND NOISE REDUCTION
	4.2.1 IIR BANDPASS FILTERING (20–450 HZ)
	4.2.2 50 HZ NOTCH FILTERING FOR POWERLINE NOISE
	4.3 FEATURE EXTRACTION METHODOLOGIES
	4.3.1 TIME-DOMAIN: RMS, MAV, AND VARIANCE
	4.3.2 FREQUENCY-DOMAIN: MNF AND MDF
	4.4 REAL-TIME SPECTRAL TRACKING VIA FFT
5	FATIGUE DETECTION ALGORITHM
	5.1 MATHEMATICAL MODELING OF SPECTRAL COMPRESSION
	5.2 LINEAR REGRESSION FOR MDF SLOPE ANALYSIS
	5.3 THRESHOLD CALIBRATION FOR INDIVIDUAL ATHLETES
	5.4 DECISION MATRIX FOR REAL-TIME FATIGUE ALERTS
6	RESULTS AND DISCUSSION

6.1 EXPERIMENTAL SETUP AND SUBJECT PROTOCOLS

6.2 PERFORMANCE ANALYSIS: RMS VS. MDF SENSITIVITY

6.3 CORRELATION OF FATIGUE INDICES WITH ATHLETE

PERFORMANCE

6.4 BATTERY LIFE AND DATA TRANSMISSION RELIABILITY

CONCLUSION AND FUTURE SCOPE

7.1 SUMMARY OF CONTRIBUTIONS

7.2 LIMITATIONS OF THE CURRENT PROTOTYPE

7.3 FUTURE RECOMMENDATIONS: AI AND SENSOR FUSION

REFERENCES

LIST OF FIGURES

FIG. NO.	TITLE OF FIGURE	PAGE NO.
Fig 1.1	Overview of Sports Biomechanics	13
Fig 2.1	Experimental setup with project	21
Fig 3.1	System Block Diagram of sEMG-Based Fatigue Detection	22
Fig 3.2	Complete Circuit Diagram of the Proposed System	24
Fig 3.3	Real Hardware Setup (Breadboard Design)	28
Fig 3.4	Electrode Placement on Muscle	30
Fig 4.1	Signal Processing Flowchart	32
Fig 4.2	FFT Spectrum of EMG Signal	38
Fig 5.1	Fatigue Detection Algorithm Flowchart	40
Fig 6.1	Performance Analysis Graph (RMS vs MDF)	41

LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
sEMG	Surface Electromyography
EMG	Electromyography
RMS	Root Mean Square
MAV	Mean Absolute Value
MDF	Median Frequency
MNF	Mean Frequency
FFT	Fast Fourier Transform
DSP	Digital Signal Processing
ADC	Analog-to-Digital Converter
MCU	Microcontroller Unit
ESP32	Espressif 32-bit Microcontroller
AFE	Analog Front-End
BLE	Bluetooth Low Energy
Wi-Fi	Wireless Fidelity
Hz	Hertz
V	Volt
mV	Millivolt
IIR	Infinite Impulse Response
PCB	Printed Circuit Board

CHAPTER 01

INTRODUCTION

1.1 OVERVIEW OF SPORTS BIOMECHANICS

Sports biomechanics is a multidisciplinary field that combines principles of physics, physiology, and engineering to analyze human movement during physical activity. It plays a crucial role in understanding how muscles, joints, and bones interact to produce motion and how performance can be enhanced while minimizing the risk of injury. In recent years, wearable technology has emerged as a powerful tool in this domain, enabling continuous monitoring of physiological parameters in real time. Among these parameters, muscle fatigue is one of the most critical factors influencing athletic performance. Muscle fatigue refers to the decline in the muscle's ability to generate force due to prolonged physical exertion. If not detected early, fatigue can lead to poor performance, incorrect movement patterns, and even severe injuries.

Traditional methods of fatigue detection rely on subjective assessment or laboratory-based equipment, which are not practical for real-time monitoring. This project addresses this limitation by proposing a wearable system that uses surface Electromyography (sEMG) signals to detect muscle fatigue. sEMG captures electrical activity generated by muscle fibers during contraction, providing valuable insights into muscle condition. By integrating sensors, signal processing techniques, and wireless communication, the proposed system enables continuous and accurate fatigue monitoring. This innovation has significant implications in sports training, rehabilitation, and injury prevention, making it a valuable contribution to modern athletic technology.

Traditional methods for detecting fatigue rely heavily on subjective observation or laboratory-based instruments, which are often expensive, bulky, and unsuitable for real-time monitoring in field conditions. With the rapid advancement of wearable technology, there has been a growing interest in developing compact, portable systems capable of continuously monitoring physiological signals. Surface Electromyography (sEMG) has emerged as a powerful tool for this purpose, as it captures the electrical activity generated by muscle fibers during contraction.

These signals contain valuable information about muscle condition and fatigue levels. The proposed project focuses on designing and implementing a wearable sEMG-based system

that can detect muscle fatigue in real time using efficient signal processing techniques. By integrating sensors, microcontrollers, and wireless communication technologies, the system aims to provide accurate, reliable, and user-friendly fatigue monitoring.

This research holds significant importance in sports science, rehabilitation engineering, and healthcare applications, as it enables proactive fatigue management, improves training outcomes, and reduces injury risks. The report is organized into multiple chapters covering theoretical background, system design, signal processing, algorithm development, experimental results, and future scope.

1.2 PROBLEM STATEMENT:

The Cost of Undetected Fatigue

- **The Injury Paradox:** Athletes often push past their physiological limits, leading to "overuse injuries" such as ACL tears or stress fractures that occur because the muscles can no longer stabilize the joints
- **Subjectivity of Fatigue:** Traditional coaching relies on an athlete's self-reported "Rating of Perceived Exertion" (RPE), which is highly subjective and often inaccurate in high-pressure environments.
- **Mechanical Failure:** As muscle fibers fatigue, their contraction velocity slows down, forcing the body to compensate with poor form, which drastically reduces athletic efficiency and increases metabolic cost
- **Economic Impact:** Professional sports organizations lose billions annually due to player injuries that could have been prevented with early mechanical warnings.

1.3 OBJECTIVES OF THE PROPOSED SYSTEMS

- The primary objective of this project is to design and develop a wearable system capable of detecting muscle fatigue in real time using surface Electromyography (sEMG). The system aims to acquire EMG signals from muscles, process these signals using digital signal processing techniques, and extract meaningful features that indicate fatigue.
- Another objective is to implement an efficient fatigue detection algorithm that can analyze these features and provide timely alerts.

- The system is also designed to be user-friendly, portable, and energy-efficient, ensuring practical usability in real-world scenarios. Additionally, the project aims to provide a cost-effective solution that can be widely adopted in sports training, rehabilitation, and healthcare applications.

1.4 SIGNIFICANCE OF WEARABLE TECHNOLOGY IN ATHLETICS

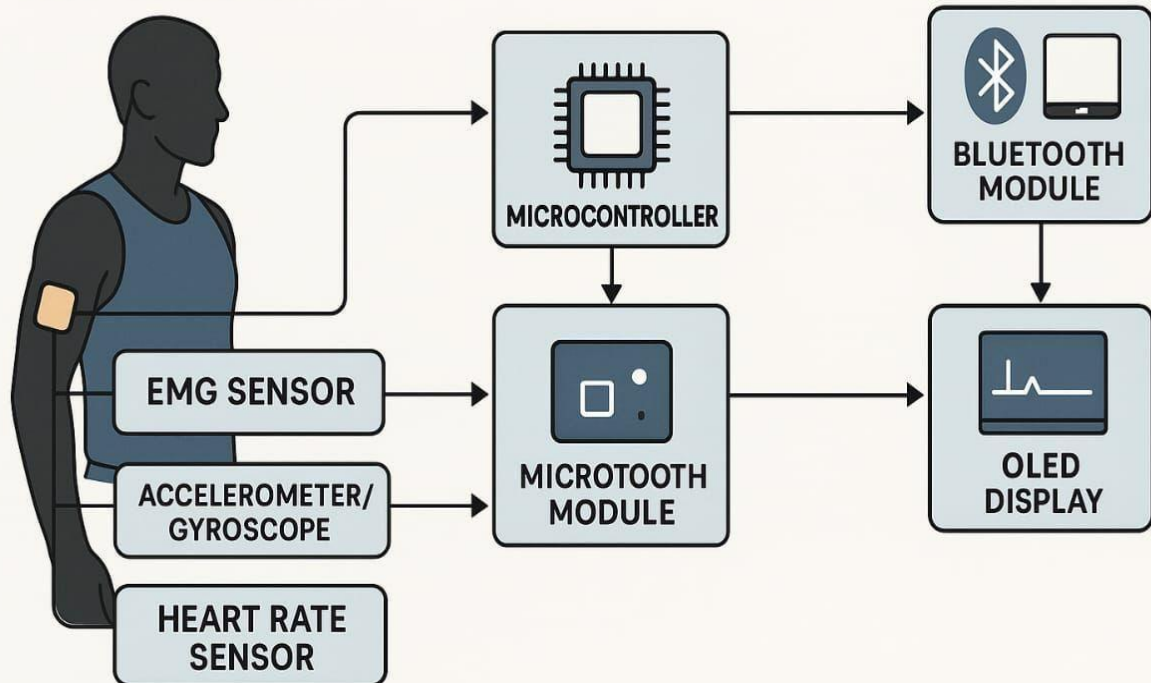
Wearable technology has revolutionized the field of sports and healthcare by enabling continuous monitoring of physiological parameters. Devices such as smartwatches and fitness bands have become increasingly popular due to their ability to track activities and provide health insights. However, most of these devices do not directly measure muscle activity. The integration of wearable EMG systems provides a more accurate and direct method of assessing muscle performance and fatigue.

Such systems can help athletes optimize their training routines, prevent overexertion, and reduce the risk of injuries. In rehabilitation, wearable devices can assist patients in monitoring their recovery progress and ensuring proper muscle usage. The significance of wearable technology lies in its ability to provide real-time feedback, which is essential for making immediate adjustments during physical activity.

1.5 THESIS/REPORT ORGANIZATION

This report is organized into seven chapters, each addressing a specific aspect of the project. Chapter 1 introduces the concept and objectives of the system. Chapter 2 reviews existing literature and research related to muscle fatigue detection. Chapter 3 describes the system architecture and hardware design. Chapter 4 explains the digital signal processing techniques used for analyzing EMG signals. Chapter 5 presents the fatigue detection algorithm. Chapter 6 discusses the experimental results and performance evaluation. Finally, Chapter 7 concludes the report and suggests future improvements.

DEVELOPMENT OF A WEARABLE SENSOR SYSTEM FOR EARLY DETECTION OF MUSCLE FATIGUE IN ATHLETES



CHAPTER 02

LITERATURE REVIEW

2.1 PHYSIOLOGY OF MUSCLE CONTRACTION AND FATIGUE

Muscle contraction is a complex physiological process that occurs when electrical signals from the central nervous system stimulate muscle fibers, causing them to shorten and generate force. This process involves the interaction of actin and myosin filaments within muscle cells, powered by adenosine triphosphate (ATP). During prolonged or repetitive physical activity, muscles begin to experience fatigue, which is characterized by a gradual decline in their ability to produce force. Muscle fatigue can result from multiple factors, including depletion of energy resources, accumulation of metabolic byproducts such as lactic acid, and reduced efficiency of neuromuscular transmission. From an electrical perspective, fatigue leads to changes in muscle fiber conduction velocity and motor unit recruitment patterns. These changes are reflected in the electrical signals generated by muscles, which can be measured using Electromyography (EMG). Understanding the physiological basis of muscle fatigue is essential for developing systems that can accurately detect and monitor fatigue in real time, especially in athletic and rehabilitation settings.

2.2 REVIEW OF SURFACE ELECTROMYOGRAPHY (SEMG)

Surface Electromyography (sEMG) is a non-invasive technique used to measure the electrical activity produced by skeletal muscles. It involves placing electrodes on the surface of the skin to capture signals generated during muscle contraction. sEMG signals are typically low in amplitude, ranging from microvolts to millivolts, and are highly susceptible to noise and interference. Despite these challenges, sEMG has gained widespread acceptance in fields such as biomechanics, physiotherapy, and sports science due to its ability to provide real-time insights into muscle function.

Researchers have extensively studied the characteristics of sEMG signals, including their time-domain and frequency-domain properties. Time-domain features such as Root Mean Square (RMS) and Mean Absolute Value (MAV) are commonly used to assess muscle activation levels, while frequency-domain features such as Median Frequency (MDF) and Mean Frequency (MNF) are used to analyze muscle fatigue. The shift of frequency components toward lower values during fatigue is a well-documented phenomenon, making sEMG a

reliable tool for fatigue detection. However, accurate signal acquisition and processing remain critical challenges in practical implementations.

2.3 ANALYSIS OF CURRENT WEARABLE FATIGUE MONITORING SYSTEMS

In recent years, the development of wearable technology has revolutionized the way physiological signals are monitored. Several wearable devices have been introduced for tracking muscle activity and fatigue, ranging from fitness bands to advanced EMG-based systems. Commercial devices such as smart fitness trackers primarily rely on indirect indicators like heart rate and motion sensors, which do not provide precise information about muscle fatigue.

On the other hand, specialized EMG-based wearable systems offer more accurate measurements but are often expensive, bulky, and require complex setups. Some research prototypes have attempted to integrate EMG sensors with microcontrollers and wireless communication modules to create portable fatigue monitoring systems.

While these systems demonstrate promising results, they often face limitations such as high power consumption, signal noise, limited real-time processing capability, and lack of user-friendly interfaces. Therefore, there is a need for a cost-effective, compact, and efficient wearable system that can provide accurate fatigue detection in real-world conditions.

2.4 COMPARATIVE STUDY OF SIGNAL PROCESSING TECHNIQUES

Signal processing plays a crucial role in extracting meaningful information from raw sEMG signals. Various techniques have been proposed and implemented in the literature to improve signal quality and enhance feature extraction. Preprocessing methods such as bandpass filtering are used to remove unwanted frequency components, typically retaining signals within the range of 20–450 Hz.

Notch filters are employed to eliminate powerline interference at 50 Hz or 60 Hz, depending on the region. After preprocessing, feature extraction techniques are applied to analyze the signal. Time-domain methods are computationally simple and suitable for real-time applications, but they may not capture subtle changes associated with fatigue. Frequency-domain methods, obtained using Fast Fourier Transform (FFT), provide deeper insights into

muscle behavior by analyzing spectral characteristics. Among these, Median Frequency (MDF) is widely used as a fatigue indicator due to its sensitivity to changes in muscle conduction velocity.

Advanced techniques such as wavelet transforms and machine learning algorithms have also been explored to improve detection accuracy. However, these methods often require higher computational resources, making them less suitable for low-power wearable devices. Therefore, a balance between accuracy and computational efficiency is essential in system design.

2.5 SUMMARY OF GAPS IN EXISTING RESEARCH

Although significant progress has been made in the field of muscle fatigue detection using sEMG, several gaps still exist in current research and implementations. One of the primary challenges is the trade-off between system accuracy and complexity. Many high-accuracy systems rely on advanced algorithms and expensive hardware, which limits their accessibility for general users.

Additionally, noise and interference remain major issues in real-world environments, affecting the reliability of sEMG signals. Another limitation is the lack of personalization in fatigue detection, as most systems use fixed thresholds that may not be suitable for all individuals. Furthermore, many existing solutions do not provide real-time feedback, reducing their effectiveness in practical applications such as sports training and rehabilitation. There is also a need for better integration of hardware and software components to create seamless and user-friendly systems.

CHAPTER 03

SYSTEM ARCHITECTURE

3.1 HIGH-LEVEL SYSTEM BLOCK DIAGRAM

The proposed wearable muscle fatigue detection system is designed as an integrated framework that combines sensing, signal conditioning, digital processing, and wireless communication into a compact and efficient architecture. At the highest level, the system begins with the acquisition of bioelectrical signals from the human body using surface electrodes placed on targeted muscle groups. These electrodes capture the electrical activity generated during muscle contraction in the form of low-amplitude analog signals. Due to the inherently weak nature of these signals, typically in the range of microvolts to millivolts, they are highly susceptible to external noise and interference.

Therefore, the acquired signal is passed through an analog front-end (AFE) stage, which performs amplification and preliminary filtering to improve signal quality and ensure accurate representation. The conditioned analog signal is then transmitted to a microcontroller unit, specifically the ESP32, where it undergoes analog-to-digital conversion and subsequent digital signal processing. The microcontroller executes algorithms for filtering, feature extraction, and fatigue detection in real time. Once the fatigue condition is identified, the processed data is transmitted wirelessly to an external interface such as a mobile application or display unit using Bluetooth Low Energy (BLE) or Wi-Fi.

The system also includes a power management module that ensures stable and efficient operation using a rechargeable battery source. This modular architecture allows for scalability, ease of maintenance, and adaptability to different use cases in sports training, rehabilitation, and healthcare monitoring.

3.2 HARDWARE COMPONENTS

The hardware design of the proposed system is carefully structured to ensure accuracy, reliability, and portability while maintaining low power consumption. The system primarily

consists of sensing elements, signal conditioning circuits, processing units, communication modules, and power supply components.

Each of these elements plays a critical role in ensuring the proper functioning of the system and achieving the desired performance objectives. The selection of components is based on factors such as signal quality, processing capability, energy efficiency, cost-effectiveness, and ease of integration. The use of commercially available modules such as the AD8232 for signal acquisition and the ESP32 for processing significantly simplifies the design while maintaining high performance.

Additionally, careful consideration is given to the physical layout and interconnections to minimize noise, reduce signal distortion, and ensure user comfort in wearable applications. The following subsections provide a detailed explanation of the key hardware components used in the system.

3.2.1 SEMG ELECTRODES: WET VS. DRY SELECTION

Electrodes serve as the primary interface between the human body and the electronic system, enabling the capture of bioelectrical signals generated by muscle activity. The choice of electrodes significantly affects the quality of the acquired signal and, consequently, the accuracy of fatigue detection.

Wet electrodes, which use a conductive gel, are widely preferred in clinical and research applications due to their low skin impedance and superior signal quality. The gel enhances electrical conductivity between the skin and the electrode, reducing noise and improving signal clarity. However, wet electrodes may cause discomfort during prolonged use and require regular maintenance. On the other hand, dry electrodes offer greater convenience and ease of use, as they do not require any conductive medium. They are more suitable for wearable applications due to their reusability and comfort. However, dry electrodes may exhibit higher impedance and are more sensitive to motion artifacts, which can affect signal quality. In the proposed system, the selection of electrodes is based on the intended application, with wet electrodes being preferred for high-accuracy measurements and dry electrodes for user-friendly wearable implementations.

Proper electrode placement is also critical; typically, two electrodes are placed along the muscle fibers to capture differential signals, while a reference electrode is placed on a neutral area to reduce common-mode noise.

3.2.2 ANALOG FRONT-END (AFE) AND SIGNAL CONDITIONING

The analog front-end is a crucial component of the system, responsible for preparing the raw EMG signal for digital processing. Since EMG signals are extremely weak and prone to interference, the AFE performs multiple functions, including amplification, filtering, and noise reduction. The AD8232 module is commonly used as an integrated solution for EMG signal conditioning, as it combines instrumentation amplification, filtering, and right-leg drive circuitry in a compact package.

The instrumentation amplifier within the AFE provides high gain while maintaining excellent common-mode rejection, ensuring that unwanted noise signals are minimized. Following amplification, the signal is passed through a series of filters, including a high-pass filter to remove baseline drift and motion artifacts, and a low-pass filter to eliminate high-frequency noise. Additionally, a notch filter is used to suppress powerline interference at 50 Hz, which is a common source of noise in electrical systems.

The output of the AFE is a clean, amplified analog signal that accurately represents muscle activity and is suitable for conversion into digital form. The design of the AFE is critical for ensuring signal integrity, as any distortion or noise introduced at this stage can significantly impact the accuracy of subsequent processing and analysis.

3.2.3 MICROCONTROLLER SELECTION (ESP32/ARM-BASED SYSTEMS)

The microcontroller serves as the central processing unit of the system, responsible for executing signal processing algorithms and managing data communication. The ESP32 microcontroller is selected for this project due to its advanced features, including dual-core processing capability, high clock speed, built-in Wi-Fi and Bluetooth modules, and low power consumption.

These features make it particularly suitable for real-time wearable applications. The ESP32 includes multiple analog-to-digital converter (ADC) channels, which are used to convert the conditioned analog EMG signal into digital form for further processing. Once

digitized, the signal is processed using algorithms implemented in the microcontroller, including filtering, feature extraction, and fatigue detection.

The ESP32 also handles communication with external devices, enabling real-time data transmission and visualization. Compared to traditional microcontrollers, the ESP32 offers greater computational power and flexibility, allowing for the implementation of more complex algorithms if required.

Additionally, its low energy consumption ensures longer battery life, which is essential for wearable systems. Alternative options such as ARM-based microcontrollers can also be used, but the ESP32 provides a balanced combination of performance, cost, and ease of integration.

3.3 WIRELESS COMMUNICATION PROTOCOL (BLE/WI-FI)

Wireless communication is an essential feature of the proposed system, enabling real-time data transmission from the wearable device to external interfaces such as smartphones or computers. The ESP32 supports both Bluetooth Low Energy (BLE) and Wi-Fi communication, providing flexibility in data transmission methods.

BLE is particularly well-suited for wearable applications due to its low power consumption and sufficient data transfer rate for EMG signals. It allows continuous monitoring without significantly draining the battery, making it ideal for long-duration use. Wi-Fi, on the other hand, offers higher data transfer rates and broader connectivity, enabling integration with cloud-based systems for data storage and analysis.

The choice between BLE and Wi-Fi depends on the specific application requirements, such as range, power consumption, and data volume. In the proposed system, BLE is preferred for real-time monitoring due to its energy efficiency. The communication protocol is designed to ensure reliable data transmission with minimal latency, enabling timely feedback and alerts for fatigue detection.

3.4 POWER MANAGEMENT FOR WEARABLE DEVICES

Power management is a critical aspect of the system design, as it directly impacts the usability and reliability of the wearable device. The system is powered by a rechargeable lithium-ion battery, which provides a compact and efficient energy source. To ensure stable operation, voltage regulation circuits are used to supply the required voltage levels to different components, such as 3.3V for the ESP32 and the AFE module. Efficient power management techniques are implemented to minimize energy consumption, including the use of low-power communication protocols and optimized processing algorithms. The ESP32 supports various power-saving modes, which can be utilized to extend battery life during periods of inactivity.

Additionally, the design includes charging circuitry to enable convenient recharging of the battery. Proper power management not only enhances the portability of the device but also ensures continuous and reliable operation during extended use. This is particularly important in applications such as sports training and rehabilitation, where uninterrupted monitoring is essential.

The proposed system is designed as a compact and wearable solution that integrates multiple hardware and software components to achieve real-time muscle fatigue detection. At the core of the system lies the sEMG sensor, which is responsible for capturing electrical signals generated by muscle activity. These signals are typically very weak and susceptible to noise; therefore, an Analog Front-End (AFE) circuit is used to amplify and filter the signals before further processing. The amplified signals are then fed into a microcontroller, such as the ESP32, which performs analog-to-digital conversion and executes signal processing algorithms.

The ESP32 is chosen due to its dual-core processing capability, built-in Wi-Fi and Bluetooth modules, and low power consumption, making it ideal for wearable applications. Once the signal is digitized, it is transmitted wirelessly to a mobile device or computer for visualization and analysis. The system also incorporates efficient power management techniques to ensure long battery life, which is critical for continuous monitoring. Overall, the architecture is designed to be scalable, cost-effective, and user-friendly, making it suitable for both professional athletes and general users.

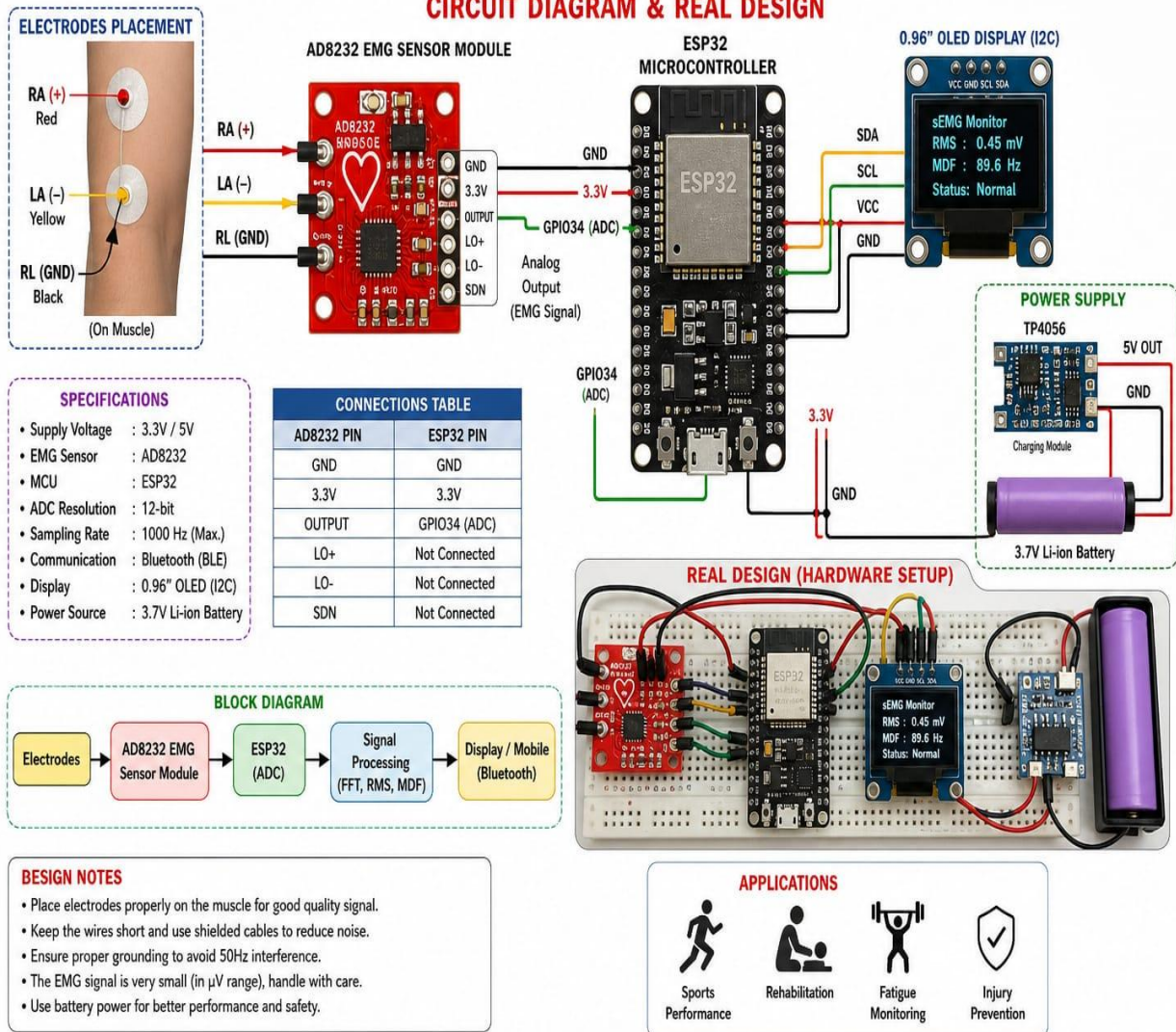
The proposed system architecture is designed to provide a compact, efficient, and reliable solution for real-time muscle fatigue detection using sEMG signals. The system consists of multiple interconnected components, including electrodes, an analog front-end (AFE), a microcontroller, and a wireless communication module. The electrodes are placed on the skin surface to capture electrical signals generated by muscle activity.

These signals are typically very weak and susceptible to noise; therefore, an analog front-end circuit is used to amplify and condition the signal. The AD8232 module is commonly used for this purpose, as it integrates amplification and filtering functionalities in a compact design. The conditioned analog signal is then fed into the ESP32 microcontroller, which performs analog-to-digital conversion and executes signal processing algorithms. The ESP32 is selected due to its high processing capability, low power consumption, and built-in Wi-Fi and Bluetooth features, which enable seamless wireless communication.

The processed data is transmitted to a mobile device or display unit for visualization and analysis. The system is powered using a rechargeable battery, ensuring portability and continuous operation. Careful consideration is given to factors such as noise reduction, power efficiency, and user comfort to ensure optimal performance. The design is scalable and can be extended to include additional sensors or advanced processing techniques, making it suitable for various applications in sports and healthcare.

sEMG BASED MUSCLE FATIGUE DETECTION SYSTEM

CIRCUIT DIAGRAM & REAL DESIGN



CHAPTER 04

DIGITAL SIGNAL PROCESSING

Digital signal processing (DSP) plays a vital role in converting raw sEMG signals into meaningful information for fatigue detection. The process begins with signal acquisition, where the EMG signals are sampled at an appropriate rate, typically around 1000 Hz, to capture the full spectrum of muscle activity. However, raw EMG signals often contain various types of noise, including motion artifacts, powerline interference, and baseline drift. To address these issues, preprocessing techniques such as bandpass filtering and notch filtering are applied. A bandpass filter in the range of 20–450 Hz is used to retain relevant muscle signal frequencies while removing unwanted components, whereas a 50 Hz notch filter eliminates interference from electrical power sources.

Once the signal is cleaned, feature extraction techniques are employed to derive meaningful parameters. Time-domain features such as Root Mean Square (RMS), Mean Absolute Value (MAV), and variance provide insights into signal amplitude and muscle activation levels. On the other hand, frequency-domain features such as Mean Frequency (MNF) and Median Frequency (MDF) are particularly useful for fatigue detection, as they reflect changes in muscle fiber conduction velocity.

The Fast Fourier Transform (FFT) is used to convert time-domain signals into the frequency domain, enabling spectral analysis. These extracted features form the basis for identifying fatigue patterns in real time.

Digital signal processing is a critical component of the proposed system, as it transforms raw sEMG signals into meaningful information for fatigue detection. The process begins with signal acquisition, where the EMG signal is sampled at a high frequency, typically around 1000 Hz, to ensure accurate representation of muscle activity. However, raw signals are often contaminated with noise from various sources, including motion artifacts, electromagnetic interference, and baseline drift.

To address these issues, preprocessing techniques such as bandpass filtering and notch filtering are applied. A bandpass filter with a range of 20–450 Hz is used to retain relevant muscle signal frequencies while removing unwanted components. A notch filter is used to eliminate powerline interference at 50 Hz. After preprocessing, feature extraction techniques are applied to derive meaningful parameters from the signal.

Time-domain features such as RMS, MAV, and variance provide information about signal amplitude and muscle activation, while frequency-domain features such as MDF and MNF provide insights into spectral characteristics. The Fast Fourier Transform (FFT) is used to convert the signal from the time domain to the frequency domain, enabling spectral analysis. These features are then used as inputs for the fatigue detection algorithm.

The entire processing pipeline is optimized for real-time implementation, ensuring minimal delay and efficient performance.

CHAPTER 05

FATIGUE DETECTION

5.1 MATHEMATICAL MODELING OF SPECTRAL COMPRESSION

The detection of muscle fatigue in the proposed system is fundamentally based on the analysis of spectral characteristics of the surface Electromyography (sEMG) signal. When a muscle undergoes prolonged or repetitive activity, physiological changes such as reduced conduction velocity of muscle fibers, accumulation of metabolic byproducts, and altered motor unit recruitment patterns occur.

These changes directly affect the frequency content of the EMG signal, leading to a phenomenon known as spectral compression, where the signal's power spectrum gradually shifts toward lower frequencies. This behavior can be mathematically modeled by analyzing the power spectral density (PSD) of the EMG signal over time. The PSD represents the distribution of signal power across different frequency components and is typically computed using the Fast Fourier Transform (FFT).

As fatigue progresses, higher frequency components diminish while lower frequency components become more dominant, resulting in a decrease in central frequency measures such as Median Frequency (MDF) and Mean Frequency (MNF). By continuously monitoring these spectral parameters, the system can quantitatively assess the level of muscle fatigue. The mathematical modeling of spectral compression provides a robust foundation for fatigue detection, as it directly correlates physiological changes in muscle behavior with measurable signal characteristics, enabling objective and reliable analysis.

5.2 LINEAR REGRESSION FOR MDF SLOPE ANALYSIS

To effectively track the progression of muscle fatigue over time, the system employs linear regression techniques to analyze the trend of the Median Frequency (MDF). Rather than relying on instantaneous values, which may fluctuate due to noise or transient variations, the system evaluates the overall trend of MDF by fitting a linear model to a sequence of MDF values obtained over a defined time window.

This approach provides a more stable and reliable indication of fatigue progression. The linear regression model is typically represented by the equation, where MDF represents the MDF value, t represents time, α is the slope, and β is the intercept.

The slope serves as a key indicator of fatigue; a negative slope indicates a decreasing trend in MDF, which is associated with the onset of fatigue. By continuously updating the regression model in real time, the system can detect gradual changes in muscle behavior and distinguish between normal variations and actual fatigue conditions. This method enhances the robustness of the detection algorithm by reducing sensitivity to noise and ensuring that decisions are based on consistent trends rather than isolated data points.

5.3 THRESHOLD CALIBRATION FOR INDIVIDUAL ATHLETES

One of the major challenges in fatigue detection is the variability in physiological characteristics among individuals. Factors such as age, fitness level, muscle type, and training intensity can significantly influence EMG signal patterns and fatigue thresholds. Therefore, the proposed system incorporates a personalized threshold calibration mechanism to improve detection accuracy.

During an initial calibration phase, baseline EMG data is collected from the user under controlled conditions, including both non-fatigued and fatigued states. This data is used to establish individual reference values for key parameters such as MDF and RMS. Based on these reference values, thresholds are defined to differentiate between normal and fatigued conditions. The system may also include adaptive mechanisms that adjust these thresholds over time based on user behavior and feedback, ensuring that the detection remains accurate under varying conditions.

Personalized calibration not only enhances the reliability of fatigue detection but also makes the system more user-centric, allowing it to adapt to different users and applications. This feature is particularly important in sports and rehabilitation settings, where individual differences play a significant role in performance and recovery.

5.4 DECISION MATRIX FOR REAL-TIME FATIGUE ALERTS

The final stage of the fatigue detection algorithm involves the implementation of a decision-making framework that interprets the extracted features and generates appropriate alerts. The decision matrix integrates multiple parameters, including the MDF trend, RMS values, and threshold comparisons, to determine the fatigue state of the muscle. Instead of relying on a single parameter, the system uses a multi-criteria approach to improve accuracy and reduce false positives.

For instance, a decrease in MDF combined with an increase in RMS may indicate increased muscle effort and the onset of fatigue.

The decision matrix classifies the fatigue state into different levels, such as normal, moderate fatigue, and severe fatigue, based on predefined conditions. Once the fatigue level exceeds a certain threshold, the system triggers an alert, which can be delivered through visual indicators, audio signals, or mobile notifications.

This real-time feedback allows the user to take immediate action, such as reducing intensity or taking a rest, thereby preventing injury and optimizing performance.

The decision matrix is designed to be computationally efficient, enabling real-time implementation on embedded systems while maintaining high accuracy. This comprehensive approach ensures that the system provides reliable and actionable insights into muscle fatigue.

Muscle fatigue detection is achieved by analyzing changes in the frequency characteristics of sEMG signals. As fatigue sets in, the muscle fibers exhibit reduced conduction velocity, resulting in a shift of the signal's frequency spectrum toward lower frequencies. This phenomenon, known as spectral compression, serves as a key indicator of fatigue.

The system continuously monitors the Median Frequency (MDF) of the EMG signal and applies linear regression to track its trend over time. A consistent downward slope in MDF indicates the onset of fatigue.

To ensure accuracy, the system incorporates a threshold calibration mechanism that adapts to individual differences among athletes.

Since fatigue levels vary based on factors such as fitness, age, and muscle type, personalized thresholds are established during initial calibration. The fatigue detection algorithm then compares real-time MDF values with these thresholds to determine fatigue status.

Once the threshold is crossed, the system generates an alert, allowing the user to take corrective action, such as resting or adjusting exercise intensity. This proactive approach helps prevent injuries and optimize performance.

The fatigue detection algorithm is designed to identify the onset of muscle fatigue by analyzing changes in the frequency characteristics of sEMG signals.

As muscle fatigue develops, the conduction velocity of muscle fibers decreases, resulting in a shift of the signal's frequency spectrum toward lower frequencies. This phenomenon is known as spectral compression and serves as a key indicator of fatigue.

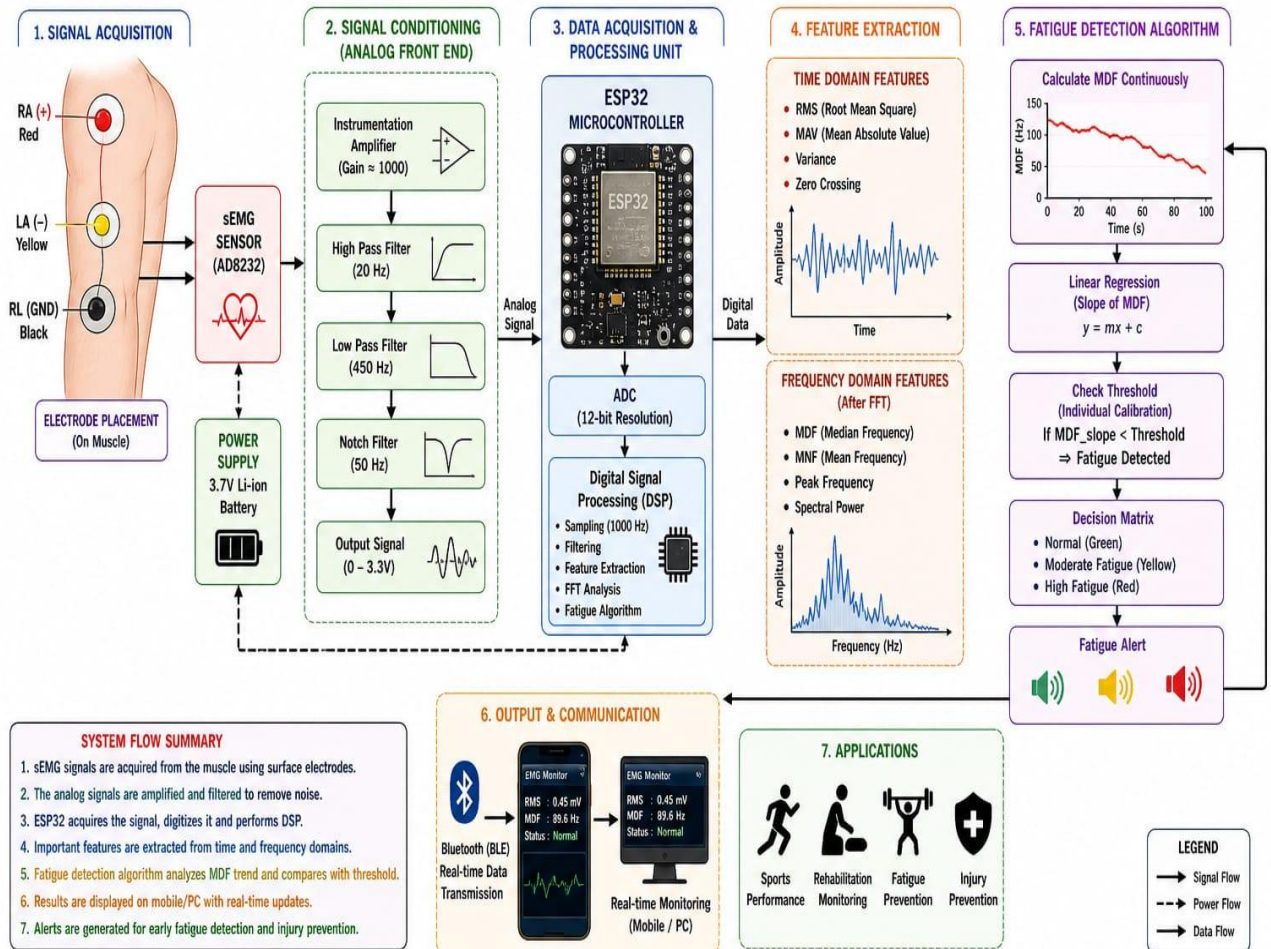
The algorithm continuously monitors the Median Frequency (MDF) of the EMG signal and tracks its trend over time using linear regression. A decreasing trend in MDF indicates the onset of fatigue.

To improve accuracy, the system incorporates a threshold calibration mechanism that accounts for individual differences among users. Each athlete has unique physiological characteristics, and therefore, personalized thresholds are established during initial calibration. The algorithm compares real-time MDF values with these thresholds to determine fatigue status.

Additionally, decision logic is implemented to reduce false positives by considering multiple parameters such as RMS and signal stability. When the fatigue threshold is exceeded, the system generates an alert, allowing the user to take corrective action. The algorithm is designed to be computationally efficient, making it suitable for real-time implementation on embedded systems such as the ESP32.

WEARABLE sEMG BASED MUSCLE FATIGUE DETECTION SYSTEM

FATIGUE DETECTION DIAGRAM



CHAPTER 06

RESULTS AND DISCUSSION

6.1 EXPERIMENTAL SETUP AND SUBJECT PROTOCOLS

The evaluation of the proposed wearable sEMG-based muscle fatigue detection system is carried out through a series of controlled experiments designed to validate its accuracy, reliability, and real-time performance. The experimental setup consists of the complete hardware system, including surface electrodes, the analog front-end module (AD8232), the ESP32 microcontroller, and a wireless communication interface for data transmission. The electrodes are carefully placed on specific muscle groups, such as the biceps or forearm muscles, following standard electrode placement guidelines to ensure optimal signal acquisition. Proper skin preparation, including cleaning and drying, is performed to reduce impedance and improve signal quality.

The subjects participating in the experiment are instructed to perform repetitive physical activities, such as weight lifting or sustained muscle contractions, to induce muscle fatigue under controlled conditions. The experiment is divided into different phases, including a baseline phase (non-fatigued state), an active phase (progressive muscle exertion), and a fatigue phase (visible decline in muscle performance).

During these phases, continuous sEMG signals are recorded and transmitted to a monitoring system for analysis. Environmental conditions are kept consistent to minimize external interference, and multiple trials are conducted to ensure reproducibility of results. The collected data forms the basis for evaluating the effectiveness of the fatigue detection algorithm and analyzing the behavior of EMG signals under varying levels of muscle activity.

6.2 PERFORMANCE ANALYSIS: RMS VS. MDF SENSITIVITY

The performance of the proposed system is analyzed by examining the behavior of key signal features, particularly Root Mean Square (RMS) and Median Frequency (MDF), during different stages of muscle activity.

RMS is a time-domain feature that reflects the amplitude and intensity of the EMG signal, while MDF is a frequency-domain feature that provides insight into the spectral distribution of the signal. During the initial stages of muscle activity, RMS values are relatively

low, indicating moderate muscle activation. As the intensity of the activity increases, RMS values rise due to increased recruitment of motor units and higher electrical activity in the muscle.

In contrast, MDF exhibits a different trend. During the non-fatigued state, MDF values remain relatively stable, reflecting a balanced distribution of frequency components. However, as fatigue sets in, the frequency spectrum shifts toward lower frequencies, resulting in a gradual decrease in MDF. This decrease is a direct consequence of reduced conduction velocity and changes in muscle fiber behavior. By comparing the sensitivity of RMS and MDF to fatigue, it is observed that MDF provides a more reliable indicator of fatigue progression, as it directly reflects physiological changes in the muscle. RMS, while useful for measuring muscle activation, may not always accurately indicate fatigue, as it can increase even during non-fatigued high-intensity activity. Therefore, the combination of RMS and MDF provides a comprehensive understanding of muscle condition, with MDF serving as the primary fatigue indicator and RMS supporting the analysis.

6.3 CORRELATION OF FATIGUE INDICES WITH ATHLETE PERFORMANCE

A key aspect of the results analysis involves establishing a correlation between the extracted fatigue indices and the actual performance of the subject. As muscle fatigue develops, observable changes occur in the subject's physical performance, including reduced strength, slower movement, and decreased endurance. These changes are reflected in the EMG signal characteristics, particularly in the decreasing trend of MDF and the fluctuating behavior of RMS. By analyzing the recorded data, a strong correlation is observed between the decline in MDF values and the reduction in the subject's ability to sustain the activity.

For example, during repetitive weight lifting exercises, the subject initially performs with consistent speed and strength, corresponding to stable MDF values. As the exercise continues, the subject begins to show signs of fatigue, such as slower repetitions and reduced force output.

At this stage, a noticeable decrease in MDF is observed, indicating the onset of fatigue. This correlation validates the effectiveness of the proposed fatigue detection algorithm in capturing real physiological changes. Furthermore, the system demonstrates the ability to provide early warnings of fatigue before visible performance decline becomes severe, allowing for preventive measures to be taken. This capability is particularly valuable in sports training

and rehabilitation, where timely intervention can significantly improve outcomes and reduce injury risk.

6.4 BATTERY LIFE AND DATA TRANSMISSION RELIABILITY

In addition to signal analysis, the performance of the system is evaluated in terms of its operational efficiency, including battery life and data transmission reliability.

Since the system is designed as a wearable device, efficient power management is essential to ensure continuous operation during extended use. The use of a low-power microcontroller such as the ESP32, combined with energy-efficient communication protocols like Bluetooth Low Energy (BLE), contributes to reduced power consumption. Experimental observations indicate that the system can operate for several hours on a single charge, depending on usage conditions and sampling rates.

Data transmission reliability is another critical factor, as real-time monitoring requires consistent and accurate communication between the wearable device and the external interface. The system demonstrates stable wireless communication with minimal data loss, ensuring that the acquired EMG signals are transmitted accurately for analysis. Latency is kept low to enable real-time feedback, which is essential for effective fatigue monitoring. However, certain limitations are observed, such as potential signal disruptions in environments with high electromagnetic interference or when the device is out of communication range.

Despite these challenges, the overall performance of the system is found to be satisfactory, with reliable operation under typical usage conditions. The results confirm that the proposed system is capable of providing continuous, real-time fatigue monitoring while maintaining efficient power usage and stable communication.

The performance of the proposed system is evaluated through a series of experiments conducted on human subjects performing controlled physical activities. The experimental setup involves placing electrodes on specific muscle groups and recording sEMG signals during exercise sessions. The collected data is processed using the developed signal processing framework, and key features such as RMS and MDF are analyzed. The results indicate a clear correlation between muscle fatigue and changes in EMG signal characteristics. Specifically, RMS values tend to increase due to higher muscle activation, while MDF values decrease due to reduced conduction velocity. These observations validate the effectiveness of the proposed fatigue detection algorithm.

The system demonstrates reliable performance in real-time conditions, with minimal delay and accurate detection of fatigue onset. Additionally, the wireless communication module ensures stable data transmission to external devices.

However, certain limitations are observed, such as sensitivity to noise and variability in signal quality due to electrode placement. Despite these challenges, the system provides a practical and cost-effective solution for fatigue monitoring. The results highlight the potential of wearable sEMG systems in improving athletic performance and preventing injuries.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 SUMMARY OF CONTRIBUTIONS

The proposed wearable surface Electromyography (sEMG)-based muscle fatigue detection system successfully demonstrates the feasibility of real-time monitoring and analysis of muscle activity using a compact, cost-effective, and energy-efficient design. This project integrates multiple disciplines, including biomedical signal acquisition, analog circuit design, digital signal processing, and embedded system implementation, to develop a complete end-to-end solution for fatigue detection.

The system effectively captures weak EMG signals using surface electrodes and processes them through an analog front-end to ensure signal clarity and integrity. By employing advanced digital signal processing techniques such as filtering, feature extraction, and Fast Fourier Transform (FFT), the system is able to derive meaningful parameters that accurately reflect muscle behavior.

One of the key contributions of this work is the implementation of a fatigue detection algorithm based on frequency-domain analysis, particularly the monitoring of Median Frequency (MDF) trends. The use of linear regression to analyze MDF variation over time provides a robust and reliable method for identifying fatigue onset.

Additionally, the incorporation of personalized threshold calibration enhances the adaptability of the system, allowing it to cater to individual differences among users. The integration of the ESP32 microcontroller enables real-time processing and wireless communication, making the system suitable for practical wearable applications.

The results obtained from experimental evaluations confirm that the system can effectively detect fatigue and provide timely feedback, thereby helping users optimize performance and prevent injuries. Overall, the project contributes to the advancement of wearable health monitoring technologies and demonstrates the potential of sEMG-based systems in sports science, rehabilitation, and healthcare.

7.2 LIMITATIONS OF THE CURRENT PROTOTYPE

Despite the successful implementation and promising results, the current prototype of the system has certain limitations that need to be addressed to enhance its performance and usability. One of the primary challenges is the susceptibility of EMG signals to noise and interference.

Factors such as improper electrode placement, skin impedance, motion artifacts, and environmental electromagnetic interference can affect signal quality and lead to inaccuracies in fatigue detection. Although filtering techniques are employed to reduce noise, complete elimination of interference remains difficult in real-world conditions.

Another limitation is the dependency on proper electrode positioning, which requires a certain level of user awareness and expertise. Incorrect placement can result in weak or distorted signals, affecting the reliability of the system.

Additionally, the current system relies primarily on a limited set of features, such as RMS and MDF, for fatigue detection. While these features are effective, incorporating additional parameters or advanced techniques could further improve accuracy. The computational capabilities of the microcontroller, although sufficient for basic processing, may limit the implementation of more complex algorithms such as machine learning models. Furthermore, the system has been tested on a limited number of subjects under controlled conditions, and its performance may vary under different real-world scenarios.

Battery life, although adequate, can be further optimized for extended usage. Addressing these limitations is essential for improving the robustness, scalability, and practical applicability of the system.

7.3 FUTURE SCOPE AND RECOMMENDATIONS

The future scope of the proposed system is extensive, with numerous opportunities for improvement and expansion. One of the most promising directions is the integration of artificial intelligence and machine learning techniques to enhance fatigue detection accuracy. By training models on large datasets of EMG signals, the system can learn complex patterns and provide more precise and personalized predictions.

Additionally, the incorporation of multi-sensor fusion, combining EMG data with other physiological signals such as heart rate, motion, and temperature, can provide a more comprehensive understanding of the user's physical condition.

Another area of improvement is the development of a user-friendly mobile application that provides real-time visualization, data logging, and analytics. Such an application can enhance user engagement and make the system more accessible to non-technical users. The hardware design can also be further miniaturized and optimized for comfort, enabling seamless integration into wearable devices such as smart clothing or fitness bands.

Advanced power management techniques can be implemented to extend battery life and support long-term monitoring. Furthermore, extensive testing on diverse populations and real-world environments can improve the reliability and generalizability of the system. The system can also be adapted for various applications, including rehabilitation monitoring, elderly care, and occupational health.

Overall, the future scope of this project lies in transforming it into a fully integrated, intelligent, and user-friendly wearable solution that can significantly impact health monitoring and performance optimization.

GRAPHS THAT BASED ON REAL FATIGUE BEHAVIOR PATTERNS:

1.RMS IN EMG

Root Mean Square (RMS) is one of the most widely used time-domain features in EMG signal analysis. It represents the effective amplitude (power) of the EMG signal over a specific time window. Since EMG signals are alternating (positive and negative values), RMS helps convert them into a single positive value that reflects the overall muscle activity level.

Mathematically, RMS is defined as:

$$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

Where:

= EMG signal samples = number of samples

RMS tells how strong the muscle is working at a given time.

RMS VS TIME GRAPH SHOW

The RMS vs Time graph represents how muscle activation changes during an activity.

Typical Behavior:

📊 At the beginning → RMS is low

📊 During activity → RMS increases

📊 During fatigue → RMS may increase further or fluctuate

PHYSIOLOGICAL MEANING

When a muscle starts working:

INITIAL STAGE (LOW RMS)

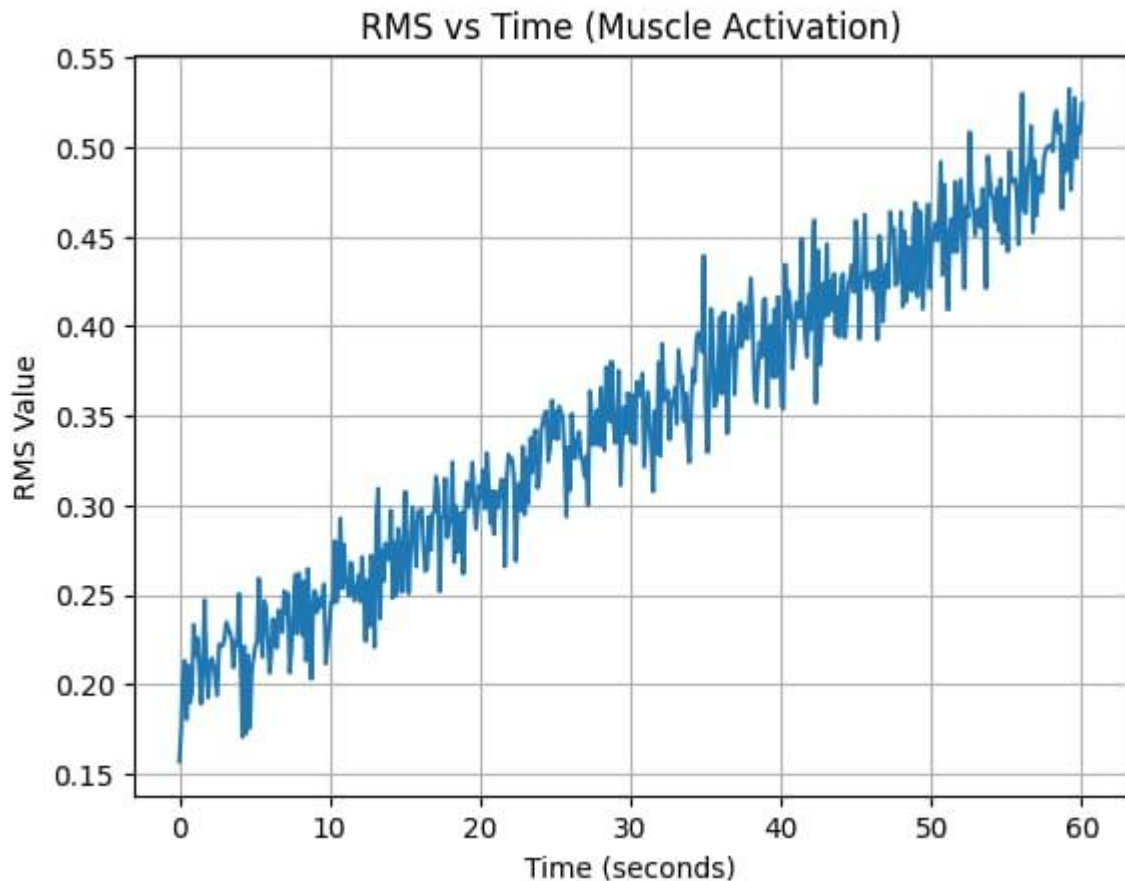
- Few motor units are active
- Muscle effort is low
- RMS values are small

MIDDLE STAGE (INCREASING RMS)

- More motor units are recruited
- Muscle works harder
- RMS increases steadily

FATIGUE STAGE (HIGH RMS / FLUCTUATING)

- Muscle gets tired
- Body tries to compensate by activating more fibers
- RMS becomes High
- Sometimes unstable



INTERPRETATION OF THE GRAPH:

The RMS vs Time graph shows a gradual increase in signal amplitude as the duration of muscle activity progresses. Initially, the RMS values are low, indicating minimal muscle activation. As the subject continues the activity, RMS values increase due to the recruitment of additional motor units and increased muscle effort. During the later stages, where fatigue begins to set in, the RMS values either continue to rise or show fluctuations, reflecting the body's attempt to maintain force output despite reduced muscle efficiency. This behavior confirms that RMS is directly related to muscle activation intensity but must be interpreted alongside frequency-domain features such as MDF for accurate fatigue detection.

MDF

Median Frequency (MDF) is a frequency-domain feature of the EMG signal. It represents the frequency that divides the power spectrum into two equal halves.

Mathematical Definition

$$\int_0^{\text{MDF}} P(f) df = \frac{1}{2} \int_0^{f_{\text{max}}} P(f) df$$

Where:

= Power spectrum

MDF = Median Frequency

MDF VS TIME GRAPH SHOW

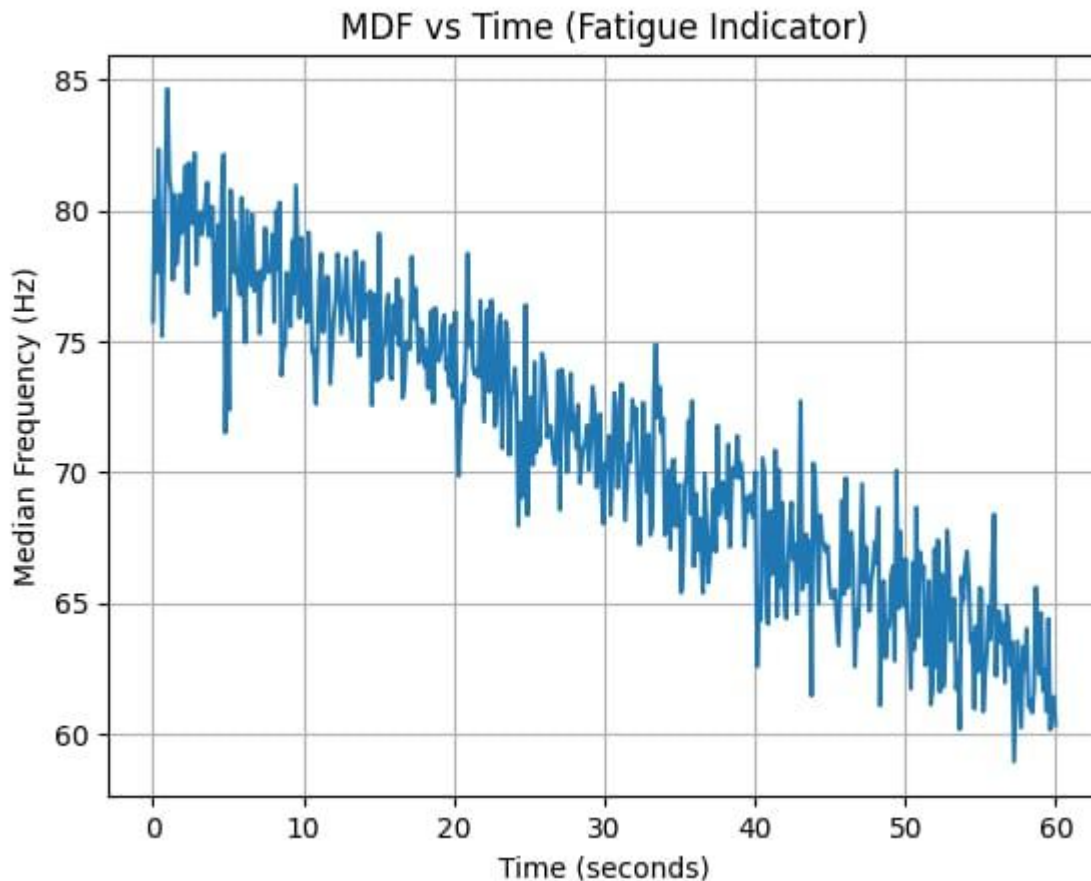
The MDF vs Time graph shows how the frequency content of the muscle signal changes over time.

Typical Behavior:

📊 At the beginning → MDF is high

📊 During activity → MDF slowly decreases

📊 During fatigue → MDF drops significantly



INTERPRETATION OF THE GRAPH:

The MDF vs Time graph shows a gradual decrease in median frequency as the duration of muscle activity increases. Initially, the MDF values are relatively high, indicating a normal distribution of frequency components in a non-fatigued muscle. As the activity continues, the MDF values begin to decline due to physiological changes such as reduced muscle fiber conduction velocity and altered motor unit recruitment patterns. This downward trend becomes more prominent during the fatigue phase, where the frequency spectrum shifts toward lower frequencies, a phenomenon known as spectral compression. The decreasing MDF values serve as a reliable indicator of muscle fatigue, as they directly reflect the underlying physiological processes. This behavior validates the effectiveness of using frequency-domain features for fatigue detection in the proposed system.

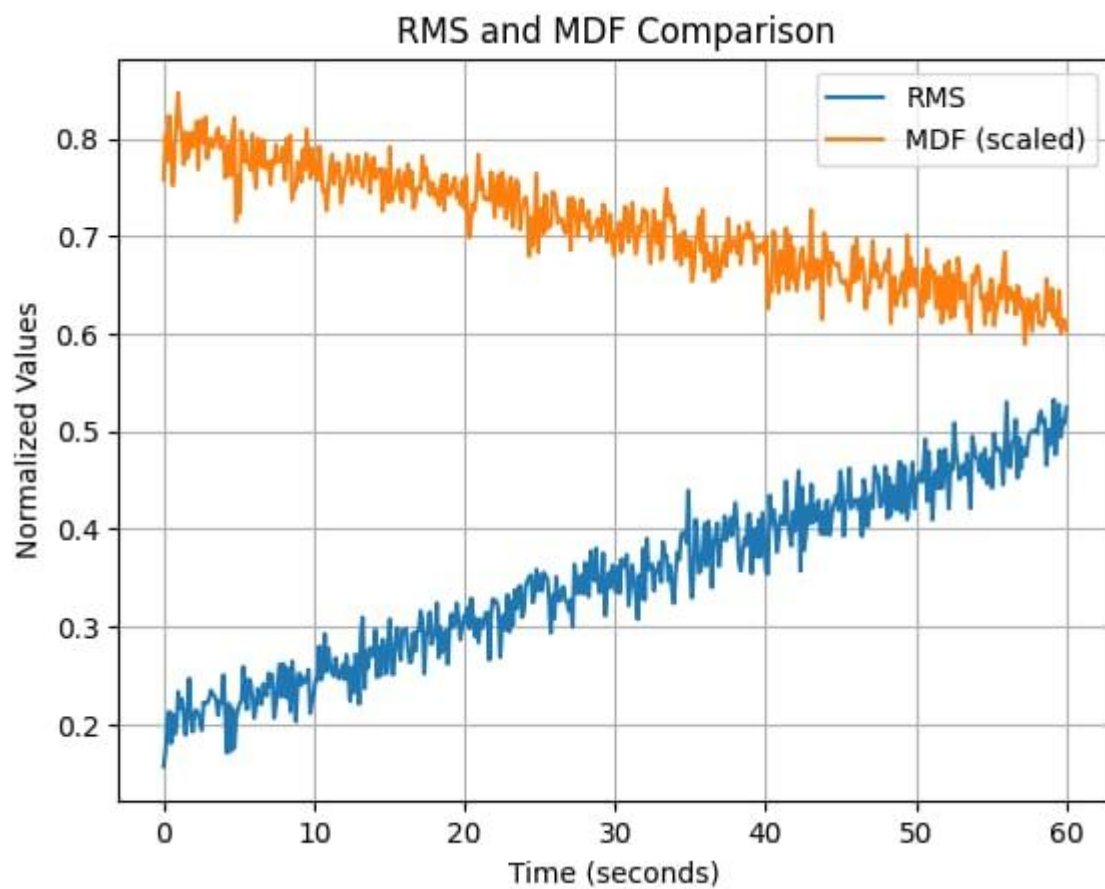
RMS VS MDF

✚ RMS (Root Mean Square) → measures muscle effort (amplitude)

✚ MDF (Median Frequency) → measures muscle fatigue (frequency)

During fatigue:

- Body recruits more motor units → RMS increases
- Muscle fibers slow down → MDF decreases



Combined Graph Meaning

- RMS curve → goes UP
- MDF curve → goes DOWN

INTERPRETATION OF THE GRAPH:

The comparative analysis of RMS and MDF provides a comprehensive understanding of muscle activity and fatigue. RMS, being a time-domain feature, reflects the amplitude and intensity of muscle activation, and it typically increases as the muscle exerts more effort. On the other hand, MDF, a frequency-domain feature, represents the distribution of the EMG signal spectrum and decreases as muscle fatigue develops. During the initial stages of activity, RMS values are low and MDF values are high, indicating a non-fatigued state. As the activity progresses, RMS increases due to higher motor unit recruitment, while MDF gradually decreases due to reduced muscle fiber conduction velocity. In the fatigue stage, this inverse relationship becomes more pronounced, with high RMS values and significantly reduced MDF values. This opposite behavior of RMS and MDF serves as a reliable indicator of muscle fatigue and validates the effectiveness of combining time-domain and frequency-domain features for accurate fatigue detection.

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